

Abhirup Saha

📍 Kolkata, West Bengal, India

✉ abhirup5002@gmail.com

☎ +91 76858 80749

🐙 GitHub profile

🌐 LinkedIn Profile

Education

Indian Institute of Engineering Science and Technology, Shibpur

Aug, 2023 – June, 2027

B.Tech in Computer Science and Technology; CGPA: **9.56/10** (upto 6th Semester)

Howrah, WB, India

Kendriya Vidyalaya No.2, Saltlake

April, 2023

CBSE; AISSCE (12th) Score: **95%**

Saltlake, WB, India

Kendriya Vidyalaya No.2, Saltlake

April, 2021

CBSE; AISSE (10th) Score: **96.4%**

Saltlake, WB, India

Work Experience

RESEARCH INTERN at VECC — Nuclear Pulse Denoising 📄

May, 2026 – July, 2026

- Architected a 4-stage physics-guided **Variational Autoencoder (VAE-PGNN)** for nuclear pulse denoising, achieving a **10× noise reduction** and **+22 dB SNR**.
- Synthesized large-scale training data via **Monte Carlo** simulations and benchmarked against classical filters optimized with **meta-heuristic** algorithms.

Projects

SmartDiner: AI-Powered Restaurant Assistant 📄

August, 2026 – Present

- Engineered a governed **multi-stage AI pipeline (FastAPI, Next.js 15, PostgreSQL)** using **Gemini 3.5 Flash Lite** with an **ILP solver (PuLP)** to optimize meals across **6** restaurants and **104** dishes, guaranteeing **100%** allergen safety.
- Built real-time **Server-Sent Events (SSE)** streaming, a **Multi-Venue Concierge** resolver, and an **Admin** dashboard with AI insights and analytics integration, validated by **15** PyTest suites including LLM accuracy and E2E pipeline tests.
- Architected a **Dual-Tier in-memory cache** (L1 Catalog + L2 Constraint Filter) with **SQLAlchemy 2.0 (asyncio)** and **UUIDv7**, achieving **sub-millisecond** refinement latency and eliminating N+1 queries.
- Deployed backend on **Render** (Dockerized) and frontend on **Vercel**, with **WORM** audit logging to **Google Cloud Storage** and **GitHub Actions** CI for automated wiki sync.

Scanity: AI-Powered Document Q&A System 📄

September, 2026 - Present

- Built an enterprise-grade **RAG** system (**FastAPI, Next.js 15, PostgreSQL 16 + pgvector**) with decoupled **Celery + Redis** workers for async PDF ingestion and **768-dim** vector embedding generation.
- Implemented **HNSW** approximate nearest-neighbor indexing for **sub-5ms** cosine similarity retrieval, with an anti-hallucination gate rejecting queries below **0.70** similarity threshold.
- Developed dual-layer conversation history (client-side **localStorage** + durable **PostgreSQL** audit tables), auto-titling sessions, and a glassmorphic chat UI with verified citation popovers.

Beyond Rules: Adaptive Fraud Detection for Banks 📄

June, 2025 – September, 2025

- Developed a **multi-layered AI fusion** system to detect banking fraud, analyzing over **10,000** real-time transactions per second using **FastAPI** and **PyTorch**.
- Integrated an on-device **TFLite** behavioral model and a **Graph Neural Network (GNN)** utilizing **Neo4j** to perform dual-validation, reducing false positives by **40%**.
- Built a cross-platform **Flutter** application for banking clients, decreasing fraud detection latency by **35%** while detecting complex anomalous patterns.

Shelly: Custom UNIX-based Shell 📄

August, 2026

- Developed a custom UNIX shell in **C** utilizing POSIX system calls (**fork, execvp, wait**) for process creation, handling over **50+** concurrent background jobs efficiently.
- Implemented robust parsing and handling for both internal built-in commands and external system binaries, reducing command execution overhead by **15%**.
- Engineered advanced shell capabilities including input/output redirection and signal handling, leveraging **GCC** and **Make** to automate build processes across **3** UNIX environments.

Notely: Microservices-based Note-taking Platform

April, 2026 – August, 2026

- Designed a robust **microservices architecture** using an **Express API gateway**, supporting up to **5,000** concurrent users with independent authentication, user, and note services.
- Containerized the entire backend infrastructure using **Docker** and **Docker Compose**, improving deployment speed by **50%** with polyglot persistence utilizing **PostgreSQL, MongoDB, and Redis**.
- Developed a responsive Single-Page Application (SPA) using **React**, served via **Nginx**, achieving a **99.9%** uptime and **<200ms** API response times.

SMB Client App: Android app to access SMB file-shares

February, 2026 – August, 2026

- Engineered a **Flutter (Dart)** Android application for accessing **SMBv2/3** network shares, implementing NTLM authentication and hierarchical file navigation for **10,000+** files.
- Secured user credentials with hardware-backed **AES-256 GCM** encryption via Android Keystore and mitigated MITM risks through strict local subnet IP validation with **0%** data leakage.
- Optimized file operations by building an asynchronous multi-file download manager with in-memory **ZIP compression**, reducing storage footprint by **30%** and cutting download times by **25%**.

SpectraBias: Media Alignment & Bias Detection

April, 2025 - May, 2025

- Engineered a scalable **web scraper** leveraging **Selenium, BeautifulSoup**, and **parallel processing** to compile and sanitize a curated dataset of over **1,300** Indian news articles.
- Fine-tuned a **BigBird-RoBERTa** transformer model to classify long-form political text into ideological leanings, optimizing a weighted **F-beta score** (1.8) to counteract severe class imbalance.
- Deployed a full-stack **FastAPI** web application to serve real-time predictions via URL or raw text input, integrating **Hugging Face** tokenizers and **PyTorch** inference engines.

IPL Data Analysis and Winner Prediction

March, 2025

- Engineered a predictive ML pipeline leveraging a **StackingClassifier** (RandomForest & SVC), fine-tuned via **hyperopt** to forecast IPL match winners with **79.71% accuracy** and **79.99% F1-score**.
- Conducted comprehensive **EDA** and **feature engineering** on **17 seasons** of IPL data, resolving data inconsistencies and deriving **5 new analytical datasets** to map team and player performance trends.
- Designed and evaluated an **Artificial Neural Network (ANN)** using **PyTorch** with **SiLU** activations alongside classical ensembles, demonstrating the superiority of ensemble methods on smaller tabular datasets.

QR On-Site: QR Code Scanner and Generator

November, 2025 – March, 2026

- Developed a lightweight, browser-based QR code scanner and generator using **JavaScript, HTML5**, and **CSS3**, processing **50+** QR formats natively.
- Integrated client-side cryptography and web APIs to perform instantaneous decoding and rendering of QR payloads in under **100ms** entirely on the user's device, ensuring **100%** data privacy.

Technical Skills

Languages: Python, C/C++, TypeScript, SQL, HTML/CSS, Dart

Frameworks & AI: PyTorch, FastAPI, Next.js, React, Generative AI (RAG), Prompt Engineering

Cloud & DevOps: GCP, Terraform, Docker, GitHub Actions, PyTest

Databases & Tools: PostgreSQL, MongoDB, Redis, Neo4j, Git, Linux

Core Subjects: OOPS, CN, DBMS, OS

Achievements

- **Finalist** in **IIESTxUCO Bank Hackathon** on **Fraud Detection in Banking Transactions**
- Ranked **1st** among 29 teams in **Truth Tuner, 2025**, building an **ML model** for Indian News Bias Detection

Team Experience

LEAD . CORE MEMBER — **Cybersecurity Chapter : CodeIIEST, GDGoC IIESTS** October 2025 – Present

- Successfully organized **RootAccess CTF, EOFool CTF** in AY 2025-26 with **300+** participants.
- Conducted **4+** technical workshops on **Cryptography** and **Computer Networks**, mentoring **100+** students across departments.